

Grab System

Grab and throw objects in Unity.

V 1.1.5

- Incomplete documentation parts will be improved over time.
- Get the most up to date documentation by [clicking here](#).
- Remember you can hover over fields in the “Inspector” window in Unity’s editor to read tooltip explanations of each field.
- If you have any questions or need assistance email support at intuitivegamingsolutions@gmail.com.

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- A module that provides useful helper functions and tools for collisions, rigidbodies, velocity tracking and more..
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- A component that allows you to make grabbing easy by allowing ConditionalGrabber grabbers to automatically grab a GrabbableObject while the ConditionalGrabber grabs while in the grab area trigger.

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- A component that allows ignore collisions between all colliders in a Grabber and some specified colliders while the Grabber is grabbing the relevant GrabbableObject.

5.c. [The DelayUnignoreCollidersOnRelease Component](#)

- A component that works hand-in-hand with the IgnoreCollidersWhileGrabbing component to allow you to specify a delay for unignoring collisions on release of a grabbable object.

5.d. [The MaintainOffsetByGrabOffset Component](#)

- A component that allows different 'maintain offset' Transforms to be specified based on the Grabber's offset relative to the GrabbableObject at the time of grab.

5.e. [The MaintainOffsetByProjectedAngle Component](#)

- A component that allows different 'maintain offset' Transforms to be specified based on the Grabber's angle around some local axis relative to the GrabbableObject at the time of grab.

5.f. [The MaintainOffsetByRelativeAngle Component](#)

- A component that allows different 'maintain offset' Transforms to be specified based on the Grabber's relative angle to the GrabbableObject at the time of grab..

5.g. [The GrabberReleaseArea Component](#)

- A component that can force a Grabber to release whatever it is grabbing when it enters the area and/or prevent new grabs.

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NOTE: See 'API Reference.pdf' ([online](#)) if you are looking for source code documentation.

Grabbers

2.a. The Grabber Component

- Documentation coming soon!

2.b. The RayGrabber Component

- Documentation coming soon!

2.b. The DistanceGrabber Component

- A component that provides the core functionality for making [GrabbableObjects](#) grabbable from a distance. Attached to the same [GameObject](#) as your [Grabber](#).
- Documentation coming soon!

GrabbableObjects

3.a. The GrabbableObject Component

3.a.i. The Component

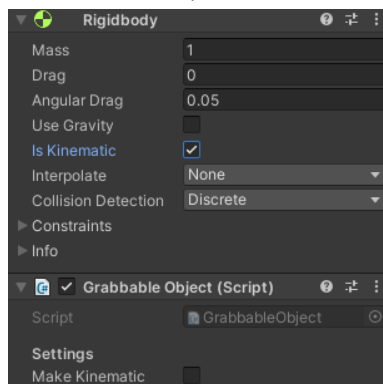
- The [GrabbableObject](#) component is responsible for making any object, with or without a [Rigidbody](#) component grabbable by any [Grabber](#).

3.a.ii. Making a 'Pinned' Grabbable (Kinematic)

- Make sure your GrabbableObject component is using the [GrabbableObject.GrabMode.MoveObject](#) grab mode.



- To make a 'pinned' [GrabbableObject](#) that stays in whatever position and rotation it is released in you will need to make a [GrabbableObject](#) that has a kinematic [Rigidbody](#) component **and** has [GrabbableObject.makeKinematic](#) disabled (as this would undo the kinematic state of the grabbable on release.)



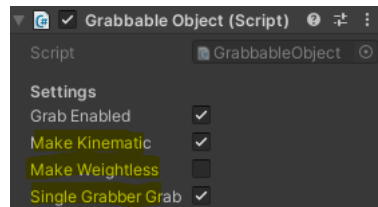
A screenshot showing the described settings where the GrabbableObject has 'make kinematic' disabled and the Rigidbody has 'is kinematic' enabled already.

3.a.iii. Making a 'Weightless' Grabbable

- Make sure your GrabbableObject component is using the [GrabbableObject.GrabMode.MoveObject](#) grab mode.

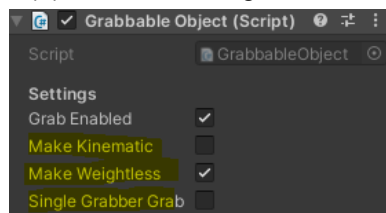


- **Single Hand Grab - Weightless Grabbable**
 - To make a 'weightless' [GrabbableObject](#) that may only be grabbed by a **single hand** you will want to ensure that the [GrabbableObject](#): (1) has a [Rigidbody](#) component, and (2) has 'Make Kinematic' enabled.



A screenshot showing which settings should be enabled and disabled..

- This will ensure that all physics properties of the grabbable object are ignored while the object is being grabbed.
 - This differs from making a 'pinned' grabbable in that the 'Make Kinematic' setting will automatically restore the non-kinematic state of the grabbable object once it is released whereas the steps from the 'pinned' grabbable section create a grabbable object that is always kinematic, even after being released.
 - ***If you do not have 'single hand grab' enabled then you will have made an ['averaged grabbable'](#) instead of a weightless single-hand grabbable.***
- **Multi Hand Grab - Weightless Grabbable**
 - To make a multi-hand grab 'weightless' [GrabbableObject](#) you will want to ensure that the [GrabbableObject](#): (1) has a [Rigidbody](#) component, (2) has 'Make Kinematic' **disabled**, and (3) has 'Make Weightless' enabled.

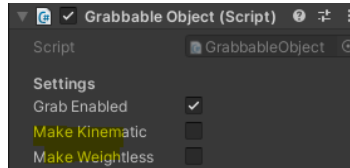


A screenshot showing which settings should be enabled and disabled..

- It is important to note that 'multi hand weightless grabbables' are not actually weightless, they just have their mass temporarily overridden with a smaller mass while held. This allows the grabbable to seem weightless while still remaining an active [Rigidbody](#).

3.a.iv. Making a 'Weighted' Grabbable

- To make a 'weighted' [GrabbableObject](#) you will want to ensure that the [GrabbableObject](#) both: (1) has a [Rigidbody](#) component, (2) does **not** have 'Make Kinematic' enabled, and (3) does **not** have 'Make Weightless' enabled..



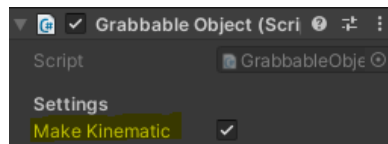
A screenshot showing which settings should be disabled..

3.a.v. Making an 'Averaged' Grabbable

- It is important to note that 'Averaged' grabbables are **rarely the correct choice**, averaged grabbables use the average position and rotation between all grabbers that are currently holding them.
- Averaged grabbables are useful for making things such as 2-handed spell effects
- Make sure your [GrabbableObject](#) component is using the [GrabbableObject.GrabMode.MoveObject](#) grab mode.



- To make an 'averaged' [GrabbableObject](#) you will want to ensure that the [GrabbableObject](#) both: (1) has a [Rigidbody](#) component, and (2) has 'Make Kinematic' enabled.

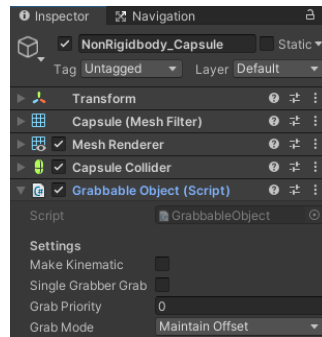


A screenshot showing which setting should be enabled..

- This will ensure that all physics properties of the grabbable object are ignored while the object is being grabbed and the position and rotation will be averaged based on all grabbers currently grabbing the grabbable..
- This differs from making a 'pinned' grabbable in that the 'Make Kinematic' setting will automatically restore the non-kinematic state of the grabbable object once it is released whereas the steps from the 'pinned' grabbable section create a grabbable object that is always kinematic, even after being released.

3.a.vi. Making a 'Stationary' Grabbable

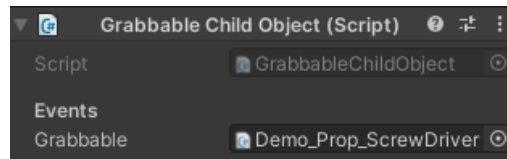
- There are many different ways to make a stationary grabbable including using the 'is kinematic' property of a [Rigidbody](#) and using the [GrabbableObject.GrabMode.MaintainOffset](#) grab mode.
- The recommended way to make a 'stationary' grabbable is to have a [GrabbableObject](#) with **no** [Rigidbody](#) component and the 'Grab Mode' set to 'Maintain Offset'.



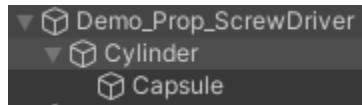
A screenshot showing a situation where the 'stationary' GrabbableObject does not have a Rigidbody component.

3.b. The GrabbableChildObject Component

- The `GrabbableChildObject` component should be attached to any child object of a `GrabbableObject` that has a `Collider` that you want to be made grabbable.
- After adding the `GrabbableChildObject` component to a child collider it should automatically find the `GrabbableObject` reference on the parent, if not make sure to manually drag the reference into the component.



A screenshot showing which setting should reference the parent GrabbableObject.



A screenshot showing what GameObject contains the GrabbableChildObject component in the hierarchy.

3.c. The DistanceGrabbable Component

- A component that makes a [GrabbableObject](#) instantaneously grabbable from a distance by a [DistanceGrabber](#). Attached to the same [GameObject](#) as your [GrabbableObject](#).
- Documentation coming soon!

3.d. The ForceDistanceGrabbable Component

- A component that makes a [GrabbableObject](#) grabbable via physics forces (or instantaneously) from a distance by a [DistanceGrabber](#). Attached to the same [GameObject](#) as your [GrabbableObject](#).
- Documentation coming soon!

3.e. The DistanceCatchAssist Component

- A component that assists a [DistanceGrabber](#) (by reference, not attached to the same GameObject) in making catching easier.
- The '**snap grab**' setting makes it so a large trigger area can be used and still have partially believable catching but can lead to clipping.
- When not using snap grab the component will briefly force a grab when the grabbable hits the trigger giving an extra moment to complete the grab manually.
- Documentation coming soon!

Modules

4.a. TimeSystem Module

- The module that provides the current time.
- Why does this module exist? This module exists to allow you to easily change what 'current time' retrieval method is used. This lets you use `realtimeSinceStartup`, `unscaledTime`, or time depending on what your game requires.
- API Reference: [Time System - API Reference](#)

4.b. PhysicsTools Module

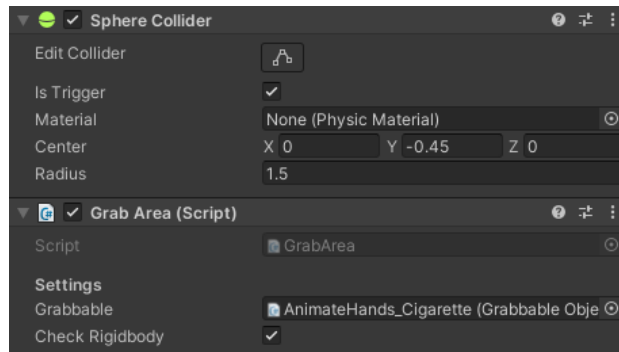
- A module that provides useful components and helper functions for many different collision-related, velocity tracking, rigidbody processing use cases and more.
- Documentation: [Physics Tools - Documentation](#)
- API Reference: [Physics Tools - API Reference](#)

4.c. PhysicsSceneManagement Module

- A module provides useful utility functions for working with `PhysicsScenes`.
- API Reference: [PhysicsSceneManagement - API Reference](#)

Extras

5.a. The GrabArea Component



- A component that allows you to make grabbing easy by allowing [ConditionalGrabber](#) grabbers to automatically grab a [GrabbableObject](#) while the [ConditionalGrabber](#) grabs while in the grab area trigger.
- The [GrabArea](#) component should be attached to the same GameObject as some **trigger Collider**. This allows a [Grabber](#) to easily grab the referenced 'Grabbable' whenever it attempts a grab in this area.

5.b. The IgnoreCollidersWhileGrabbing Component

- A component that allows ignore collisions between all colliders in a Grabber and some specified colliders while the Grabber is grabbing the relevant GrabbableObject.

5.c. The DelayUnignoreCollidersOnRelease Component

- A component that works hand-in-hand with the [IgnoreCollidersWhileGrabbing](#) component to allow you to specify an unignore collision (with grabber) delay from the time the [IgnoreCollidersWhileGrabbing](#) component was initially supposed to unignore collisions.
- Why does this component exist? Why not build it into the [IgnoreCollidersWhileGrabbing](#) component?
 - In many use cases, like in the XR gun demo, grabbable objects are deactivated once they (or other grabbable objects) are released. This would prevent delayed unignore collision events from ever occurring. To work around this behavior this component was added which can be attached to any GameObject and simply reference the relevant [IgnoreCollidersWhileGrabbing](#) component ensuring this behavior works at all times.

5.d. The MaintainOffsetByGrabOffset Component

- A component that allows different 'maintain offset' Transforms to be specified based on the [Grabber](#)'s offset relative to the [GrabbableObject](#) at the time of grab.
- This component is only useful when attached to the same [GameObject](#) as a [GrabbableObject](#) that is using the 'Maintain Offset' grab mode.

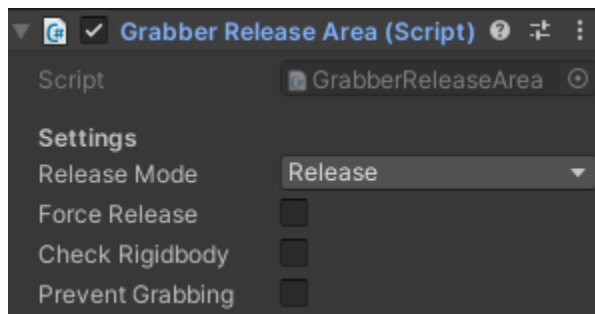
5.e. The MaintainOffsetByProjectedAngle Component

- A component that allows different 'maintain offset' Transforms to be specified based on the [Grabber](#)'s angle around some local axis relative to the [GrabbableObject](#) at the time of grab.
- This component is only useful when attached to the same [GameObject](#) as a [GrabbableObject](#) that is using the 'Maintain Offset' grab mode.

5.f. The MaintainOffsetByRelativeAngle Component

- A component that allows different 'maintain offset' Transforms to be specified based on the [Grabber](#)'s relative angle to the [GrabbableObject](#) at the time of grab.
- This component is only useful when attached to the same [GameObject](#) as a [GrabbableObject](#) that is using the 'Maintain Offset' grab mode.

5.g. The GrabberReleaseArea Component



A screenshot of the 'Inspector' pane for a GrabberReleaseArea component in v1.0.1

- A component that can force a [Grabber](#) to release whatever it is grabbing when it enters the area and/or prevent new grabs.

Setting	Description	Default Value								
Release Mode	The release mode to use.	Release <i>(GrabberReleaseArea.ReleaseMode)</i>								
	<table><tr><th>Mode</th><th>Description</th></tr><tr><td>None</td><td>No release.</td></tr><tr><td>No Throw</td><td>Release without throwing.</td></tr><tr><td>Release</td><td>Release with throwing.</td></tr></table>		Mode	Description	None	No release.	No Throw	Release without throwing.	Release	Release with throwing.
	Mode		Description							
	None		No release.							
	No Throw		Release without throwing.							
Release	Release with throwing.									
Force Release	A boolean that controls whether or not a release is forced by this area. Forced releases cannot be prevented whereas non-forced releases can be prevented through various means.	false <i>(bool)</i>								
Check Rigidbody	A boolean that controls whether or not any 'attachedRigidbody' to the triggering collider is checked for a Grabber component if none is found on the triggering Collider's gameObject.	false <i>(bool)</i>								
Prevent Grabbing	Should this area prevent Grabbers in it from grabbing?	false <i>(bool)</i>								

FAQ

(Frequently Asked Questions)

Q:

A: